MINIMALLY RELATIVISTIC SCHRÖDINGER EQUATION AND THE LINEARLY RISING POTENTIAL

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In place of the usual approximation

$$T = (m^2c^4 + p^2c^2)^{1/2} - mc^2 = p^2/2m - p^4/8m^3c^2 + O(1/c^4)$$

in the "minimally relativistic" Schrödinger equation, we suggest to employ the alternative formula

$$T = p^2c^2[(m^2c^4 + p^2c^2)^{1/2} + mc^2]^{-1} = p^2(2m + p^2/2mc^2)^{-1} + O(1/c^4).$$

Its merits are illustrated on the particular linear potentials $V(x) = \mu x +\text{const}$, where a small $O(c^{-3})$ modification of the coupling (if needed) enables us to construct the ground and first M ($= O(c^3)$) excited states $\psi(x) = e^{-cx} \times \text{polynomial}(x)$ exactly.

The phenomenological Schrödinger equation

$$[p^2/2m + V(x)]\, \psi(x) = E\psi(x), \quad p^2 = -\hbar^2 \Delta\,, \tag{1}$$

may be complemented by the various relativistic corrections (cf., e.g., § § 33 and 83 in ref. [1]). Whenever they prove to be small, eq. (1) (+ perturbations) provides a fair understanding of the whole physical system in question.

Outside of the perturbation formalisms, an inclusion of the $O(1/c^2)$ corrections may often lead to nontrivial difficulties. For example, the particular kinematic corrections

$$[p^2/2m - p^4/8m^3c^2 + V(x)]\, \psi(x) = E\psi(x) \tag{2}$$

make the exact energies complex – the new "hamiltonian" ceases to be bounded from below. In this context, our present idea lies in a transition to the "equivalent" Padé-type kinetic energy operator,

$$\left(\frac{p^2}{2m + p^2/2mc^2} + V(x)\right) \psi(x) = E\psi(x)\,. \tag{3}$$

Besides an elimination of the spurious "energy widths" in (2), the new equation (3) will exhibit also an improved asymptotic behaviour in $p \gg 1$ since the rigorous relativistic formula gives

$$(m^2c^4 + p^2c^2)^{1/2} - mc^2 \approx cp\,.$$

Furthermore, we may rewrite (3) as a second-order differential equation

$$p^2(1 + c^{-2}\, V(x) - c^{-2}E)\,\psi(x) + V(x)\psi(x) = E\psi(x),$$
$$2m = 1\,, \tag{4}$$

for a broad class of potentials, and reinterpret it finally as the "minimally relativistic" (MR) Schrödinger equation

$$p^2\chi(x) + W(x,E)\,\chi(x) = E\chi(x)\,,$$
$$\chi(x) = [1 + c^{-2}\, V(x) - c^{-2}E]\,\psi(x)\,,$$
$$W(x,E) = \frac{(1 + c^{-2}E)\, V(x) - c^{-2}E^2}{1 + c^{-2}\, V(x) - c^{-2}E}. \tag{5}$$

The original potential $V(x)$ is replaced now by its effective, weakly energy dependent modification $W(x,E)$.

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$$W(x,E) = c^2 + E - c^4/[c^2 - E + V(x)] \tag{6}$$

by definition, we may write, symbolically,

$$p^2 + c^2 = c^4/[c^2 - E + V(x_0)] > 0, \quad x_0 \in (0,\infty),$$

so that

$$E < c^2 + V(x_1), \quad x_1 \in (0,\infty).$$

This inequality resembles the Dirac discrete-spectrum restriction [1] (remember that $m = 1/2$ here). Our restriction is less restrictive for asymptotically growing potentials, but such forces are not suitable for direct use in the Klein–Gordon or Dirac equations at all. Here, eq. (6) converts such potentials into the asymptotically constant effective force $W(x,E)$ which is quite comfortable from the purely formal point of view.

As an illustrative example, let us pick up one of the most frequently used quark–antiquark interactions [2]

$$V(x) = \mu x, \tag{7}$$

and describe the corresponding semi-algebraic solution of our MR Schrödinger equation (5). Formally, after a change of variables

$$x = r^2, \quad \chi(x) = r^{1/2}\varphi(r/\sqrt{2c}), \tag{8}$$

and rescaling, we get the well-known eigenvalue problem [3]

$$[H_0 + \lambda r^2/(1+gr^2)]\varphi(r) = 0, \tag{9}$$

$$H_0 = -\mathrm{d}^2/\mathrm{d}r^2 + \mathcal{L}(\mathcal{L}+1)/r^2 + r^2,$$

$$\lambda = c^2/(E-c^2), \quad g = \mu/[2c(c^2-E)],$$

with $\mathcal{L} = 2l + 1/2$ and $l = -1, 0$ (one dimension) or $l = 0, 1, ...$ (three dimensions). In the harmonic-oscillator basis, this equation acquires the tridiagonal algebraic form

$$[(1+gT)H_0 + \lambda T]\,\varphi = 0, \tag{10}$$

$$T = \begin{pmatrix} a_0 & b_0 & & \\ b_0 & a_1 & b_1 & \\ & b_1 & a_2 & b_2 \\ & & & \ldots \end{pmatrix},$$

$$H_0 = \begin{pmatrix} 2a_0 & & \\ & 2a_1 & \\ & & \ddots \end{pmatrix}, \quad \varphi = \begin{pmatrix} \langle 0|\varphi\rangle \\ \langle 1|\varphi\rangle \\ \vdots \end{pmatrix},$$

$$a_n = 2n + \mathcal{L} + 3/2, \quad b_n = (n+1)^{1/2}(n+\mathcal{L}+3/2)^{1/2},$$
$$n = 0, 1, ...,$$

and may be solved in terms of analytic continued fractions [4].

For the particular couplings [5]

$$\lambda = -4g(M+l+1), \quad M = 0, 1, ..., \tag{11}$$

the full MR Schrödinger equation (10) degenerates, for the first $M+1$ states, to the $(M+1)$-dimensional set of equations

$$\left[H_0 - 4gT\begin{pmatrix} M+1 & & \\ & M & \\ & & \ddots \\ & & & 1 \end{pmatrix}\right]\begin{pmatrix} \langle 0|\varphi\rangle \\ \langle 1|\varphi\rangle \\ \vdots \\ \langle M|\varphi\rangle \end{pmatrix} = 0. \tag{12}$$

In the present notation, condition (11) determines simply the M-dependent "permissible" coupling constants

$$\mu = c^3/[2(M+l+1)], \tag{13}$$

while the secular equation reads

$$\det\{c^2 I - [c^2/2(M+l+1)]STS - EI\} = 0, \tag{14}$$

$$S = \begin{pmatrix} [(M+1)/a_0]^{1/2} & & & \\ & (M/a_1)^{1/2} & & \\ & & \ddots & \\ & & & (1/a_M)^{1/2} \end{pmatrix},$$

and determines therefore the first $M+1$ real binding energies in the variational-type way.

In a realistic situation with the potential $V(x) = \tilde{\mu}x$, we may satisfy (13) approximatively, with a negligible error $\tilde{\mu} - \mu = O(1/c^3)$. Then, we may also construct immediately the "low-lying" elementary solutions of the MR equation (3),

$$\psi(x) = \frac{x^{l+1}\, e^{-cx} \times \text{polynomial}(x)}{1 + \mu x/c^2 - E/c^2} \,. \tag{15}$$

In practice, these solutions will represent the whole relevant part of the spectrum since $M = O(c^3)$ is very large. In the non-relativistic limit $c \to \infty$, the indefinite character of $\psi(x)$ ($= 0 \times \infty$) is compatible also with a non-existence [6] of elementary solutions of the non-relativistic Schrödinger equation (1) with the linearly rising potential (7).

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